Krishna Tripathi

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Education

Indian Institute of Information Technology, Nagpur December 2021 to May 2025
Bachelor of Technology in Computer Science and Engineering *Nagpur, India*

Technical Skills

Languages: C++, Go, Java, Python, JavaScript, TypeScript, SQL, Rust (Basics)
Backend & Templates: Node.js, npm, Express.js, FastAPI, REST APIs, Microservices, Pug, EJS, FreeMarker
Frontend/UX: React, Next.js, CSS, SCSS, Figma **AI/LLM**: LangGraph, MCP, RAG, Langfuse
Databases: YugabyteDB, PostgreSQL, MongoDB, BigQuery, pgvector
Cloud/DevOps: AWS (S3, SQS, Lambda, Cognito), Google Cloud Run, Docker, Jenkins, GitHub Actions
Infrastructure: Kubernetes, Helm, Linux, Cloud Infrastructure, Distributed Systems

Professional Experience

Yugabyte Bengaluru, India
Software Engineer I *September 2025 to Present*

- **Replaced polling-based file ingestion with event-driven processing** using S3 ObjectCreated events, SQS, and Lambda; implemented presigned S3 uploads across 3 services with file-size, type, and spoofing checks.
- **Set up the Meko development environment** across React, Go, Python, and Nginx; configured Nginx routing for backend and Langfuse APIs.
- **Built Meko retrieval observability and context-promotion workflows** for top-k results across memory and knowledge-base sources; replaced full conversation replay, reducing input-token usage by **35%** in marketing and lead-generation workflows.
- **Implemented AWS Cognito sign-in for 3 identity providers**: Google, LinkedIn, and GitHub; added federated identity linking and profile completion, and hardened invite and logout flows.
- **Contributed AI/RAG integrations across Java and Python**: built LangChain4j embedding and chat-memory support and implemented a `PgDistRagRetriever` for pg_dist_rag. Added HNSW/hybrid search, metadata filters, and integration tests; updated documentation and contributed a merged upstream commit.
- **Improved xCluster disaster recovery with Go backend services and a React-based frontend**, reducing DR node-selection latency from **450ms to 120ms** and surfacing C++ replication-lag metrics through Prometheus for operational visibility.
- **Developed a database migration validator for the company-wide CI/CD workflow reused across 5+ repositories**; checks schema version and dirty state to block invalid migrations before merge and provide actionable feedback.
- **Strengthened YugabyteDB's open-source developer ecosystem** with JavaScript driver logging, ORM load-balancing examples, integration guidance, and JUnit tests for Yugabyted UI backend services.

Software Engineer Intern *September 2024 to August 2025*

- **Recreated the YB Voyager workflow in Yugabyted UI with UX designers using Figma** across 4 migration stages; added status and progress analytics and redesigned the migration assessment UX. (PR #2564 | implementation)
- **Built growth analytics APIs and Go workflows integrating 5 systems**: Scarf, GCS, BigQuery, MadKudu, and Salesforce to support centralized metrics, lead scoring, and CRM synchronization.
- **Streamlined yb-voyager developer tooling across 3 workflows**: added Homebrew install tracking, fixed Go/Docker setup in Jenkins helpers, standardized build-script versioning, corrected assessment HTML, and moved stable-version tests to Jenkins. (PR #2652 | PR #2779)
- **Improved YugabyteDB onboarding** by documenting TypeORM smart-driver setup and fixing Linux/RHEL quick-start links. (PR #28250 | PR #28418)

Zluri Bengaluru, India
Software Engineer Intern *August 2024 to September 2024*

- **Resolved 13+ production bugs** in JavaScript/Node.js and MongoDB; built QA dashboards and a testing framework, increasing test coverage by **20%** and improving query performance by **30%**.

Projects

EternalAgni: Agnihotra Reminder App July 2026 to Present
JavaScript, Capacitor, Android, PWA, Service Workers, Local Notifications ***SOURCE CODE*** | ***INTERNAL TESTING***

- **Built a custom NOAA timing engine** with ±**1-second** accuracy, generating sunrise and sunset schedules locally for **92 days (~3 months)** of offline use.
- **Delivered reliable reminders across Android and PWA**, supporting **2-to-60-minute** lead times with duplicate suppression, late-delivery protection, client-side scheduling, GPS/IP fallbacks, service-worker recovery, and automated APK/AAB builds.

YB Voyager Copilot: AI-Powered Database Migration Assistant January 2025 to Present
LangGraph, FastAPI, Next.js, Python, MCP, Docker, YugabyteDB, RAG ***DOCKER HUB*** | ***ARCHITECTURE*** | ***DEMO***

- **Architected a 3-agent LangGraph and MCP workflow** spanning all 4 migration stages and grounded in product documentation.
- **Built a 3-tier RAG backend and containerized 5 services** with FastAPI, pgvector, YugabyteDB, and Docker; automated image publishing through GitHub Actions and generated Playwright PDF assessments with production-sizing recommendations.

Achievements

- Received **Best Use of AI** recognition for YB Voyager Copilot at the YugabyteDB internal hackathon.
- Earned **Guardian status with a maximum rating of 2145** on LeetCode. Reached **4-star status with a 1944 rating** on CodeChef and **Specialist status with a maximum rating of 1484** on Codeforces.
- Secured **3rd place** in Algorithmia at Tantrafiesta 2023.